

- Q.29 Describe solution polymerization and mention one advantage and one disadvantage.
- Q.30 Differentiate between injection molding and extrusion molding.
- Q.31 Define degree of polymerization and write its relation with molecular weight.
- Q.32 Write any two properties and two uses of:
- Bakelite
 - Urea formaldehyde resin
- Q.33 Describe the concept of number average molecular weight.
- Q.34 What are plasticizers and stabilizers? Give one example and one function of each.
- Q.35 Explain briefly: fillers, colouring matters and cross-linking agents used as additives for plastics.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain in detail the classification of polymers (by origin, structure and intermolecular forces).
- Q.37 Describe in detail the construction and working of extrusion moulding machine with neat labelled diagram. Discuss various advantages and disadvantages.
- Q.38 Write the structure, important properties and industrial applications of:
- Polypropylene
 - Polystyrene
 - Teflon
 - Nylon 6 & Nylon 6,6

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4th Sem / Chemical Engineering Sub : Polymer Technology

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Polymer is defined as:
- Low molecular weight compound
 - Very high molecular weight compound made of repeating units
 - Simple gas
 - Ionic compound
- Q.2 The simple small molecule from which a polymer is formed is called:
- Copolymer
 - Monomer
 - Initiator
 - Plastic
- Q.3 Which of the following is a natural polymer
- Polyethylene
 - Silk
 - PVC
 - Nylon
- Q.4 Elastomers are:
- Highly elastic, rubber-like materials
 - Hard and brittle materials
 - Low molecular weight liquids
 - Glass fibers
- Q.5 A polymer formed from more than one type of monomer is called:
- Plasticizer
 - Copolymer
 - Isomer
 - All of these

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- Q.6 The degree of polymerization is:
- Number of cross links
 - Number of repeat unit in a polymer chain
 - Number of additives
 - Number of polymers mixed
- Q.7 PVC stands for:
- Polyvinyl chloride
 - Polyvinyl carbazole
 - Polyviscous chloride
 - Polyvinyl carbon
- Q.8 Bakelite is a/an
- Thermosetting polymer
 - Elastomer
 - Thermoplastic
 - None of these
- Q.9 The polymer cannot be processed again and again is known as:
- Thermosetting
 - Thermoplastic
 - Both a & b
 - None
- Q.10 The substance added to increase the flexibility of plastics is called:
- Filler
 - Stabilizer
 - Plasticizer
 - Pigment

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Write the full form of PVC.
Q.12 Name the monomer of polyethylene.

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- Q.13 Define monomer.
Q.14 Name one thermosetting resin used for electrical goods.
Q.15 Give an example of a natural polymer.
Q.16 State the role of colouring materials in plastics.
Q.17 Name the process used to make hollow plastic bottles.
Q.18 What is the main function of fillers in plastics?
Q.19 Define thermoplastics.
Q.20 Name any one elastomer.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Differentiate between thermoplastic and thermosetting polymers
Q.22 State the role of molecular weight on properties of polymers.
Q.23 Explain the properties and use of polyethylene.
Q.24 Write short note on copolymer and free radical.
Q.25 With neat and labelled diagram discuss the construction of injection molding machine.
Q.26 Differentiate between step-growth (condensation) and chain-growth (addition) polymerization.
Q.27 Explain bulk polymerization and write any two advantages and any two disadvantages.
Q.28 Expand the term PVC, write various properties and use of PVC.

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